Predicting Start-up Outcomes in Pakistan: An Integrated Econometric and Multidimensional Statistical Approach

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Praise be to Allah, Lord of the Universe

"In the name of Allah, the Most Compassionate, the Most Merciful.".

There is no deity, only Allah.

None is worthy of worship but the Lord.

The creator He is, the master He is, like unto the Lord there is none.

There is no other deity, only Allah! Only Allah! Only Allah!

DEDICATION

I want to say thanks to my beloved parents for their continuous support throughout my Bachelor's journey and special thanks to my teachers, siblings and Saliha**,** for always encouraging me through my ups and downs, assuring me that I can do it.

Most importantly, to the merciful ALLAH Almighty for consistently granting me the strength and knowledge in all our endeavors.

DECLARATION

I **<u>Ms. Zoha Ehsan</u>**, Roll No. **<u>3001-BS-STAT-22</u>**, Session 2022-2026, student of B.Sc. (Hons.) in Statistics, hereby declare that the matter printed in the report titled **"Predicting Start-up Outcomes in Pakistan: An Integrated Econometric and Multidimensional Statistical Approach"** is my own work and has not been printed, published and submitted as research work, report or publication any form in any University, Research Institution etc. in Pakistan or abroad.

Dated: <u>30-04-2026</u>

Zoha Ehsan

3001-BS-STAT-22

RESEARCH COMPLETION CERTIFICATE

Certified that the research work contained in this thesis **"Predicting Start-up Outcomes in Pakistan: An Integrated Econometric and Multidimensional Statistical Approach"** has been carried out by Ms. Zoha Ehsan, Roll No. 3001-BS-STAT-22 under my supervision during his B.Sc. (Hons.) in Statistics studies.

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Zoha Ehsan

Abstract

This study develops an integrated econometric and multidimensional statistical framework to predict startup outcomes in Pakistan, focusing on survival status and revenue performance. Using a dataset of 150 Pakistani startups founded between 2010 and 2022, the research employs different statistical methods including Cox regression, linear regression, MANOVA, t-tests, ANOVA, Kaplan-Meier analysis, and K-means cluster analysis.

The findings consistently demonstrate that negative growth is the strongest predictor of startup failure. Cox regression shows that startups with negative growth have a 15.17 times higher hazard of closure compared to those with positive growth. Revenue serves as a protective factor, with each 1% increase reducing closure hazard by 25% ($p = 0.021$). Higher funding also increases closure risk, though the effect is small ($B = 0.003$, $p = 0.029$), a finding termed the "funding paradox." The t-test results confirm that active startups grow significantly faster than closed ones regardless of funding status, with large effect sizes (Cohen's $d = 1.22$ and 1.50). Linear regression reveals that company size is the strongest predictor of revenue, with each size category increase associated with 72% rise in revenue.

Kaplan-Meier survival analysis shows that startups with positive growth maintain high survival rates, while only 30.6% of negative growth startups survive past seven years. Cluster analysis identifies two meaningful clusters: a High Growth cluster (8.7% of startups) with 69.2% survival, and a Mainstream cluster (91.3% of startups) with 61.8% survival.

The study offers clear practical guidance. Entrepreneurs should prioritize growth over fundraising. Investors should exercise caution when funding startups without positive growth. Growth is the actual dividing line between survival and failure.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In 2022, Airlift raised $109 million and was hailed as Pakistan's next big success story. One year later, it shut down completely. After almost ten years of service, Careem had to suspend its ride-hailing services in 2025. These are not just headlines. They are warnings. Something is broken in how we finance, develop, and grow startups in Pakistan. This study does not guess why. It uses data to find out.

Despite these failures, the culture of startups in Pakistan is no longer on the edge. It has become mainstream. This transition is no accident. Smartphones are spreading rapidly, and gaining access to the internet is becoming affordable even in small cities. Online payments are also easy and secure, with the help of digital payment systems like JazzCash and Easypaisa. In 2021 and 2022, more than 1 billion dollars was invested in Pakistan by venture capital. Youngsters, fed up with waiting to be offered a traditional job, do not wait but create (Invest2Innovate, 2019). A clear definition of a startup is necessary before proceeding. In this study, a startup is defined as a young, technology-enabled business venture that relies on digital technologies and has the potential for rapid growth and scalability.

However, behind this bright mask is a difficult truth. Research has revealed that about 90 percent of businesses started all around the world fail (Patel, 2015; CB Insights, 2021), and only half of all companies last beyond five years. Most of those that survive remain small and do not leave any significant inventions or employment. One-third of all startups fail because there is no need in the market to sell their product. Most failures are experienced between the second and the fifth years of operation.

The country of Pakistan, with its specific problems in the region, is an echo of this global tendency. More than 55 funded startups in the country either completely closed or drastically changed their business model in 2021-24. Careem, which ceased its core ride-hailing business in Pakistan in July 2025 after more than a decade of operation, and Airlift, which raised about $109 million before shutting down without warning in 2022, are notable failures. These are not unique incidents. They are the signs of more severe problems of the ecosystem.

What makes the situation in Pakistan even more problematic is the dramatic fall of funds. According to Invest2Innovate, startup funding in Pakistan declined by 88 percent, from $355 million in 2022 to just $43 million in 2024. By November 2024, funding had dropped further to $37 million. This sudden shortage of funding, often leaving firms without a means of survival, forces many firms to abandon their expansion plans and focus solely on surviving. To frame the situation, India is getting $7.5 billion in startup capital in 2024, while it is only $43 million in Pakistan. Instead of basing their decisions on real figures, choices on which market to join, what business model to pursue, or which company to invest in are often made based on intuition. It is not that people do not want data. This is because there is no information on Pakistani companies that is collected in a methodical way and is reliable. Lack of credible research obstructs the growth of the ecosystem (Invest2Innovate, 2024).

The good news is that corporate data is becoming more available. More accurate records are kept by investment firms, incubators, and accelerators. The Invest2Innovate Pakistan Startup Ecosystem Report 2024 is a big leap to bridge the divide in data-driven insights. Econometric and statistical tools have been developed to a level of dealing with complex, real-world data and still provide meaningful insights (Hyndman and Athanasopoulos, 2018).

Such developments are utilized in this study. I answer two simple, yet effective, questions based on real data of Pakistani companies. First, what is the difference between successful and unsuccessful startups? Second, among the ones that survive, what drives higher revenue? These are not just academic questions. They matter to the youngsters who are trying to keep up morale in the team. They are significant when an investor is deciding where to invest his money. They are significant to the legislator formulating the next technological policy.

So far, this is the first research on the outcome of startups conducted in the Department of Statistics at Government College University in Lahore. No other researcher or any other student in this department has attempted to employ an integrated econometric and multidimensional statistical approach in predicting startup success or failure. Therefore, this project forms the foundation of additional entrepreneurial analytical studies at my department.

1.2 Problem Statement

Pakistan has a growing startup ecosystem, with a very high failure rate. According to research, 90 percent of startups fail all over the world, and Pakistan is no exception. It is estimated that 9 out of 10 Pakistani businesses do not last longer than three years. Over 55 sponsored startups were either shut down or underwent significant restructuring between 2021 and 2024. For Pakistani entrepreneurs, 2022 was referred to as the "exodus year" (TechJuice, 2025).

This funding situation makes the problem even worse. Many startups are left without the capital necessary to continue functioning, expand, or switch to more profitable business models. Numerous structural issues contribute to the high failure rate. According to research on company failures worldwide, a lack of product market demand is responsible for around 34% of all startup failures. 16% of failures are caused by financial issues, and 18% are caused by team-related issues. In Pakistan, there are additional issues that raise these dangers. Since 2022, the Pakistani rupee has lost 30% of its value, and the nation continues to experience economic instability. Business viability is also threatened by unforeseen tax legislation, regulatory inefficiencies, and unpredictable internet connectivity.

Startup failures are not the sole problem. Strategic judgments, policy initiatives, and investment decisions are often based on subjective judgments instead of empirical evidence. The same mistakes are made by entrepreneurs since there are no data-driven prediction models that are trustworthy. Investors lose money on ventures that appear promising but lack fundamental strength. Policymakers struggle to develop effective support programs since they do not know what is effective in the Pakistani setting.

Also, revenue performance varies widely even among surviving startups. Whereas some generate substantial income, while others struggle to survive. The factors that predict survival may differ from the factors that predict revenue. Conversely, a company with higher revenue may still fail while one with low revenue may survive. This distinction is not given much attention in the literature that is available on Pakistani startups.

This research bridges the gap by developing an integrated econometric and multidimensional statistical framework to make predictions on the outcomes of startups in Pakistan based on actual data.

1.3 Research Aim

The primary aim of this study is to develop an integrated econometric and multidimensional statistical framework to predict startup outcomes in Pakistan, specifically focusing on survival status and revenue performance.

1.4 Research Objectives

This study pursues the following specific objectives:

i. To describe the characteristics of Pakistani startups using descriptive statistics
ii. To identify factors that significantly predict startup survival
iii. To identify factors that significantly predict startup revenue and to develop a statistical model for it.
iv. To develop a statistical model for predicting startup closure.

1.5 Research Questions

This study revolves around the following research questions:

i. What are the main characteristics of startups operating in Pakistan?
ii. Which factors significantly predict whether a startup survives or closes?
iii. Which factors significantly predict how much revenue a startup generates?
iv. Can statistical models accurately predict startup closure and revenue levels?
v. How effective are multidimensional statistical methods in analyzing startup data?

1.6 Research Hypotheses

The following hypotheses are formulated based on existing literature; these are then tested in this study:

H_1: Higher funding increases the probability of startup survival.

H_2: Higher revenue reduces the likelihood of startup closure.

H_3: Higher annual growth rates decrease the risk of failure.

H_4: Larger employee count is associated with higher revenue.

H_5: Larger company size is associated with higher revenue.

H_6: Older startups generate higher revenue than younger ones.

1.7 Data and Variables Overview

This study uses a dataset of Pakistani startups containing multiple business indicators.

Outcome Variables

1. Startup Status (Active / Closed)
2. Revenue

Predictor Variables

1. Firm Characteristics:
   i. Size
   ii. Industry
   iii. City
   iv. Time (years since founding)
2. Financial Indicators:
   i. Total Funding
   ii. Funding Rounds
3. Operational Indicators:
   i. Employee Count
   ii. Annual Growth Rate

These variables are chosen because they contain the vital components of the startup's operating ability, infrastructure, and financial standing. There are two main variables: startup status and revenue; whereas the first one determines the survivability of a startup, the second one determines its financial standing. The predictive variable helps to find out the components that play a significant role in producing such output. Environmental variables of the firm such as firm size, industry, city, and time provide information regarding the environment of the startup. Financial variables like funding and funding rounds reveal the role played by outside money in producing an outcome.

1.8 Research Gap

No previous study in the Department of Statistics at Government College University, Lahore has used an integrated econometric and multidimensional statistical approach to predict startup outcomes in Pakistan. Industry news tells us what happened in the ecosystem. It reports how much money was raised and how many startups shut down. But it does not explain why these things happened or which factors actually matter. Existing academic studies rarely look at survival and revenue together. Most focus on one or the other. Within this department, methods like Cox regression, cluster analysis, and MANOVA have not been applied to Pakistani startup data before.

This study fills these gaps by using real data from Pakistani businesses and applying statistical methods.

1.9 Scope of the Study

This study focuses exclusively on the startups operating in Pakistan. The geographic area of the study is limited to Pakistani startups. The findings may not be generalizable to startups in other countries that have different legislative, cultural, or economic factors (Autio et al., 2014). I examine two aspects of startup performance in terms of their outcomes. The former is survival, which is defined as a startup having closed or continuing to operate. The second one is financial performance, as calculated by annual income. These two distinct results can be influenced by different groups of causes. A startup might be successful and earn less, or it might earn a lot of money but not succeed (Delmar & Shane, 2004).

The analysis focuses on measurable business indicators as predictors. These involve financial aspects such as revenue, fundraising rounds, and fund amounts. Examples of operational variables include the number of employees and the rate at which they grow annually. Examples of firm characteristics are size, industry, city, and age. As a result, the research lacks secondary sources and thus fails to examine subjective factors such as customer satisfaction, team dynamics, founder motivation, or leadership quality (Davidsson, 2016).

The study is limited to quantitative analysis only. Qualitative techniques like focus groups, interviews, and case studies are not included. While qualitative research can be more helpful in understanding the reasons behind the significance of particular characteristicss, this study attempts to find statistical trends and quantifiable variables rather than investigating subjective experiences (Neuman, 2014).

The time frame of the study is the period between 2010 and 2022 (when the startups were founded), and the outcomes are monitored until 2025. This means the research only captures the early to mid-stage performances and does not follow up with the startups beyond this period. Survival and revenue trends at longer periods are not addressed in this study (Cantamessa et al., 2018). Finally, the external environmental issues of political instability, industry-specific disruptions, economic downturns, or policy changes are not considered in the study. Though they are not covered in the quantitative framework of this study, these factors are certainly influencing the results of startups.

Within these boundaries the study aims to provide concrete statistical evidence on the factors that affect the survivability and revenue of Pakistani startups.

CHAPTER 2

LITERATURE REVIEW

Bruckner et al. (2022) used a qualitative narrative tool. They studied real life cases of companies that went out of business even though they raised a lot of money. They proposed a three-tiered model of failure: individual, firms, and environment. The researchers found that having a lot of money does not guarantee success. In fact, a business can fail faster if it has more money. Companies become less cost conscious if they have abundant funds. They delay changing their business model. This insight helps explain why well-capitalized businesses (such as Airlift) fail, despite having more than $100 million in funding.

Schulte-Althoff et al. (2021) adopted a quantitative approach to study digital startups' scaling. They studied three types of startups: AI firms, platform companies and service providers. They aimed to understand the growth of revenues as firms expand their workforce. The study found that revenue did not match employee numbers i.e. sublinear scaling. Moreover, they discovered that in the case of AI startups in particular, investment grew much faster than the number of employees. That suggested that, rather than these companies being ready for growth, investors might have been overexcited about AI.

Camelo Martinez (2019) used data from 5,000 Dutch startups to create algorithms. Eight different factors were used to develop three models. One model predicted which startups would secure more than one million euros. Another model predicted which startups would have 10 or more employees. The third predicted which startups would report an annual return on investment of 20% or more. The accuracy rates of the models were 71%, 71%, and 76% respectively. This demonstrated that multidimensional methods were able to predict different dimensions of startup success.

Vallarino (2023) examined several survival models in machine learning for predicting startup failures to determine which model was the best. He looked at Survival Random Forest, DeepSurv and Kernel SVM. He first compared Kaplan-Meier models to Cox regression. The study found that Random Forest and Multi-Task Logistic Regression were more accurate in predicting survival than traditional models. However, the study also highlighted that these complex models had limitations, especially with correlated factors.

Pasayat et al. (2023) used the PRISMA model to conduct the meta-analysis of 19 articles. They used forest plots to illustrate their results. The aim of the study was to establish which factors were the best predictors of venture performance in different studies. They concluded that the business idea, market size, team size, service timeliness, market growth, and founder age were significant. This meta-analysis considered multiple studies to achieve more reliable outcomes than any single study alone.

Krishna et al. (2023) reviewed 140 studies from 1993 to 2021. They applied a review approach to examine what research has found in the past three decades on the survival of startups. This study found that most studies were based on one theoretical perspective. In addition, they found that a large proportion of the research was conducted in developed economies, such as Europe and the United States. There was very little research conducted in developing countries such as Pakistan. This confirmed the existence of a research gap.

Mullins (2014) used a conceptual analysis to study venture capital funding. This research looked at the effects of the number and timing of capital-raising rounds on success. The author found that early financing negatively impacted success. Companies receiving finance too early struggled to find product-market fit. Too much money forced the company to grow prematurely. The study advised entrepreneurs to be cautious in their financing.

Rychnovský (2018) compared logistic regression and survival analysis to predict defaults. The study validated the models by using lift and Gini coefficients. The research set out to determine which model was more accurate in predicting defaults. It was found that survival models worked better than logistic regression on ex-ante validation samples. This showed survival models were better at predicting on new data. This finding influenced the choice of Cox regression, rather than logistic regression, in this study.

Hoffman and Yeh (2018) used examples of fast-growing companies to talk about blitzscaling. They studied firms that focused on growth rather than productivity and profits. Blitzscaling is very risky, say experts. Only businesses that have clear scalability opportunities or network effects, such as social media, should blitzscale. It is merely a faster path to bankruptcy for other companies.

Monk's Hill Ventures (2024) explored a strategy called negative blitzscaling. This is where startups sell products at a loss to win market share from competitors. They examined a series of real startups who have tried this approach. A very small number were able to dominate their market. Most people couldn't. These companies have no core business that can make a profit. They just collapsed. The lesson is that building a viable business is more important than market share.

TechJuice (2025) tracked startups in Pakistan funded between 2021 and 2024. They conducted an industry reporting study to document what happened to these businesses. The study showed that over 55 startups either stopped operating or changed their business strategies. These businesses failed due to expanding too soon, a lack of market understanding and focusing on growth rather than profitability. The study called 2022 the "exodus year" for Pakistani startups due to their disappearance.

Business Recorder (2025) undertook financial reporting and analysis of Pakistani startups' funding. They examined data for 2022-2024. They found that funding had dropped by 88% from $355 million in 2022 to $43 million in 2024. By November 2024, funding was $37 million. By comparison, during the same period, India received $7.5 billion in startup investments. This striking disparity demonstrated how difficult the environment was for Pakistani businesses.

Invest2Innovate (2019) collected information and created ecosystem maps of the leading cities in Pakistan. The research considered the determinants of the growing number of startups. As it became evident during the research, the startups were experiencing the advantages of the rapid spread of mobile phones, cheap internet access, and electronic money transactions such as JazzCash and Easypaisa. However, barriers to growth were also pointed out, including lack of experienced mentoring and venture capital. The ecosystem was additionally hampered by the absence of quality research in the field.

Invest2Innovate (2024) performed a trend analysis of startup financing data for 2022-2024. It was found that the investment amount decreased by 88%, going from $355 million down to $43 million. In addition, investors were said to be cautious since they lost faith because of many failures. The idea of "growth at all costs" was gone. Successful startups made sustainable enterprises.

Ciampi and Gordini (2013) used neural network models in forecasting defaulting in small firms. Their study compared the predictive capabilities of neural network models to those of traditional methods such as logistic regression models. They showed that integrating qualitative and financial variables gave better predictions than when only financial factors were considered. It was established that relevant information was not captured when only financial factors were emphasized. A different approach needed to be adopted.

Trahms et al. (2013) conducted a study on the topic of company decline and revival. They summarized results from several studies through systematic review techniques. From their study, in order to improve its market standing, new investments should be made along with cost cutting and disposal of assets. They also found that experienced management teams and entrepreneurship mindset played important roles in reviving firms.

Bullough et al. (2014) conducted a survey among business owners from Christchurch, New Zealand after two massive earthquakes hit in 2010 and 2011. The survey examined resilience, survivability, and self-efficacy. The primary objective of this research was to find out the reasons for which some firms were able to thrive whereas others were doomed to fail. Despite the dire situation, it was found that an entrepreneur's personal perseverance positively impacted on his chances to keep the business running. Entrepreneurs had greater chances of keeping their business running when remaining optimistic and persistent.

Hair et al. (2019) published a textbook on multivariate data analysis. In this book, the assumptions and proper use of cluster analysis, discriminant analysis, and multiple regression were described. The authors emphasized that before running analyses, a researcher must make sure that the data is of good quality. Moreover, one should avoid multicollinearity, fix violations of normality and keep VIF values below five.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research design, data source, variables, descriptive statistics, and statistical techniques applied to analyze startup performance and survival in Pakistan. The methodology helps to understand the approach used in the research. This chapter explains every step of the study from how the data is gathered to which statistical tests are applied.

3.2 Research Design

The present study is quantitative in nature; it is based on an explanatory approach of the study. The data for this study is secondary in nature i.e. it will include data collected from existing sources at a particular point in time. The study covers startups that are founded from 2010 to 2022, and data would be observed till 2025.

3.3 Data Source

The dataset used in this study contains information on 150 startups functioning in Pakistan. The data is collected from publicly accessible sources like The Ministry of Planning Development & Special Initiatives, Open Data Pakistan, Crunchbase, Profit by Pakistan Today and PCMI's Pakistan B2C Ecommerce Report. All the data is structured by industry type, operational status, number of employees, growth rate, revenue, and funding information.

Most of the company registry and macroeconomic data for the report is sourced from Open Data Pakistan, a government-backed initiative focused on transparency in government datasets. The data on funding for firms in Pakistan came from business database Crunchbase. Information on the distribution of startups by sector and startups located by region came from Ministry of Planning Development & Special Initiatives.

PCMI's Pakistan B2C Ecommerce Report provided analysis of retail and e-commerce technology startups in the country. The startups included in the sample are from Karachi, Lahore, and Islamabad, which are the three main entrepreneurial cities in Pakistan. The dataset contains information on revenue, funding, employee count, annual growth, industry type, and whether the company is currently operational or not. All firms in the sample are founded between 2010 and 2022.

3.4 Variables of the Study

3.4.1 Outcome Variables

In this study there are two outcome factors. The first, startup status, is a simple binary variable. The startup is still running (Active=1) or not (Closed = 0). This variable helps answer the question of what predicts survival versus failure.

The second outcome variable is startup revenue, measured in thousand US dollars. Since revenue is very skewed (i.e., many startups earned low amounts of revenue, but a small number of startups earned very high amounts of revenue), we applied a logarithmic transformation to this variable. The logarithm of revenue (i.e., Log_Revenue) is equal to the natural logarithm of revenue plus 1. This transformation makes it easier to interpret the values of the outcome variable. Results are presented as percentage changes in the outcome variable for a one-unit change in the predictor variable while holding all other variables constant.

3.4.2 Predictor Variables

The predictor variables fall into three categories:

i. Financial indicators
ii. Operational indicators
iii. Firm characteristics.

The financial indicators include total funding in the startup in thousand $, the number of funding rounds the startup has secured and the annual growth of the startup as a percentage.

The operational indicator includes the number of employees which are working at the startup. A new binary variable called Negative_Growth is also created, which takes 1 for negative growth and 0 for either positive or zero growth.

Firm size is also noted and recorded as a numeric variable; Small=1, Medium=2 and Large =3. The industry and city in which the firm operates are also recorded.

According to most studies, a startup is considered successful if it survives for three to five years. A 3-year cut point is chosen for this study. Accordingly, a startup is considered active if it continues to exist for at least three years after its founding. Only companies created in 2022 or earlier are included in the data because startups are tracked until 2025. Time (in years) is calculated as the number of years from the founding year to 2025. This variable represents the survival time (in years) for each startup.

3.4.3 Transformed Variables

Pre-processing the data for analysis required a few transformations. First, Log_Revenue is created as the log of the revenue of the startup. We also took the natural log of the total funding plus 1 for Log_Funding. This is necessary to account for the startups that received no funding, since the log of 1 is zero. A binary variable Negative_Growth is created to track whether a startup is shrinking in size. The size category names are re-encoded into a numeric variable. The founding year of each startup is used to calculate age.

3.5 Descriptive Statistics

Descriptive statistics are calculated to summarize the data and to understand the basic characteristics of the data, before progressing to the tests.

3.5.1 Descriptive Statistics of Continuous Variables

Table 3.1 includes the following continuous variables: annual growth (percentage), revenue (in thousand US dollars), employee count, total funding (in thousand US dollars), and number of funding rounds. For each variable, the table reports the minimum, maximum, mean (or mode where indicated), standard deviation, skewness, and kurtosis. These statistics help in understanding the central tendency, dispersion, and shape of the distribution for each variable. The skewness and kurtosis values are particularly important as they indicate whether the data deviate from normality, which guides the choice of appropriate statistical techniques.

Table 3.1: Descriptive Statistics of Continuous Variables (n = 150)

Variable	Min	Max	Average	Std. Deviation	Skewness	Kurtosis
Annual Growth (%)	-96.00	243.00	Mean = 15.02	44.65	2.234	7.812
Revenue ($1000)	35.00	120,000.00	Mean = 4,593.42	10,733.71	8.692	90.859
Employee Count	1	666	Mode = 5	91.13	3.766	17.548
Total Funding ($1000)	0	282,000	Mean = 7,387.50	27,507.95	7.649	69.491
Funding Rounds	0	6	Mean = 0.78	1.15	1.909	4.210

Table 3.1 reveals several important details about the distribution of revenue and total funding of the sample startups. For revenue and total funding, skewness is greater than 7 and kurtosis greater than 69. This means that both revenue and total funding are extremely skewed, with a small percentage of startups with high values of funding and most other startups with lower funding. The large revenue range of $35,000 to $120 million additionally reveals a large diversity in the revenue of individual startups.

In addition to the above data characteristics, the number of employees also exhibits positive skewness. The highest number of employees is 666, but the mode is only 5, meaning most startups have small teams. The annual growth also exhibits significant variance, ranging from -96% to +243%. This means although some startups are growing rapidly, others are rapidly declining. Most funding rounds are zero, meaning that the majority of startups have not raised money from external investors. The data seems to support using a log transformation for funding and revenue in the regression models.

Table 3.2: Startup Status Distribution

Status	Frequency
Active	94
Closed	56
Total	150

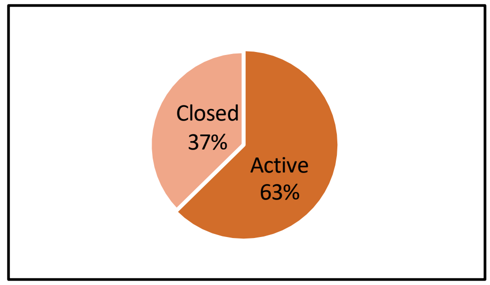


Figure: Pie chart of Startup Status

The data includes total 150 startups in which includes 94 are active ones and 56 are closed. It is visibly apparent that there are more active than closed ventures in data showing the imbalance in data. Researchers typically don't speak to founders of shut startups, because they are frequently reluctant to take part and discuss their unsuccessful projects. And in the case of failed firms, there is essentially no digital record because it vanished within two years of a company closing. Hence, where national registries maintain little or no data on closed enterprises, collating detailed information on failure is very hard.

The Pakistan Startup Ecosystem Report 2024 similarly highlights that "the lack of credible, data-driven research has been a persistent challenge for stakeholders in Pakistan's entrepreneurial ecosystem". In fact, Researchers have known for decades that they have to face the difficulty collecting enough datapoints from failed startups in emerging market contexts and same goes for Pakistan. Still, the sufficient number of closed startups (56) is enough to analyze failure patterns since Cox regression and other survival analysis methods are designed to handle imbalanced sampling schemes effectively.

3.5.2 Graphical Representation of Key Findings

To complement the descriptive statistics, four graphical representations are created to visualize the most important patterns in the data.

i. Distribution of Annual Growth Rates
ii. Startup Status by City
iii. Correlation matrix of numeric variables
iv. Average Annual Growth by size

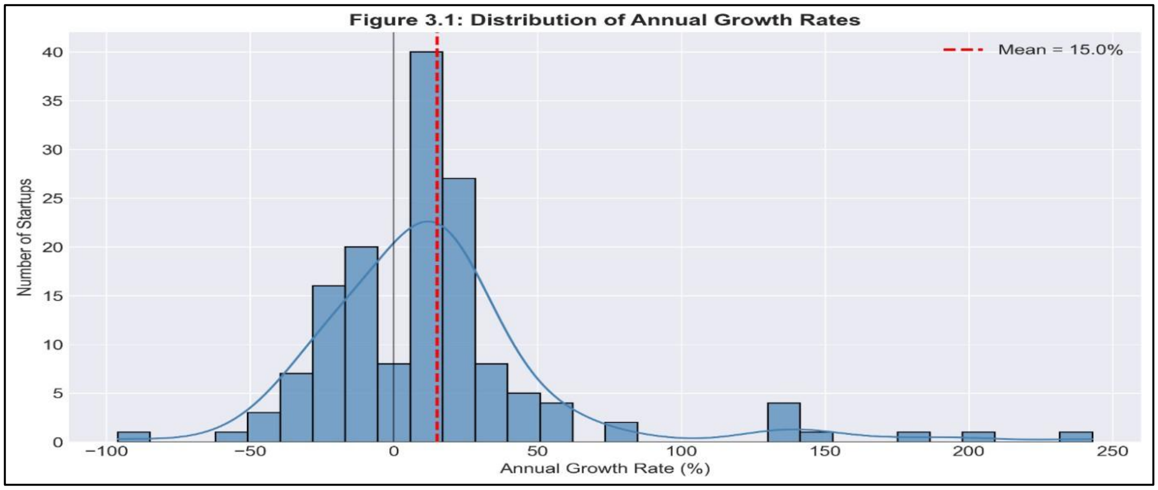


Figure 3.1: Distribution of Annual Growth Rates

Figure 3.1 shows the histogram of yearly growth rates for 150 startups. Most of them seem to be somewhere between minus 20 percent and 40 percent. There's this long tail going off to the right up to 243 percent, so the distribution is clearly right skewed. It points out a few companies that grew really fast, indicating outliers. A solid grey line sits at zero, separating the negative growth from the positive side. Then there's a red dashed line marking 15.02 percent growth, that is the average annual growth. Overall, the visualization confirms that most startups experience moderate growth, while a few achieve exceptional performance.

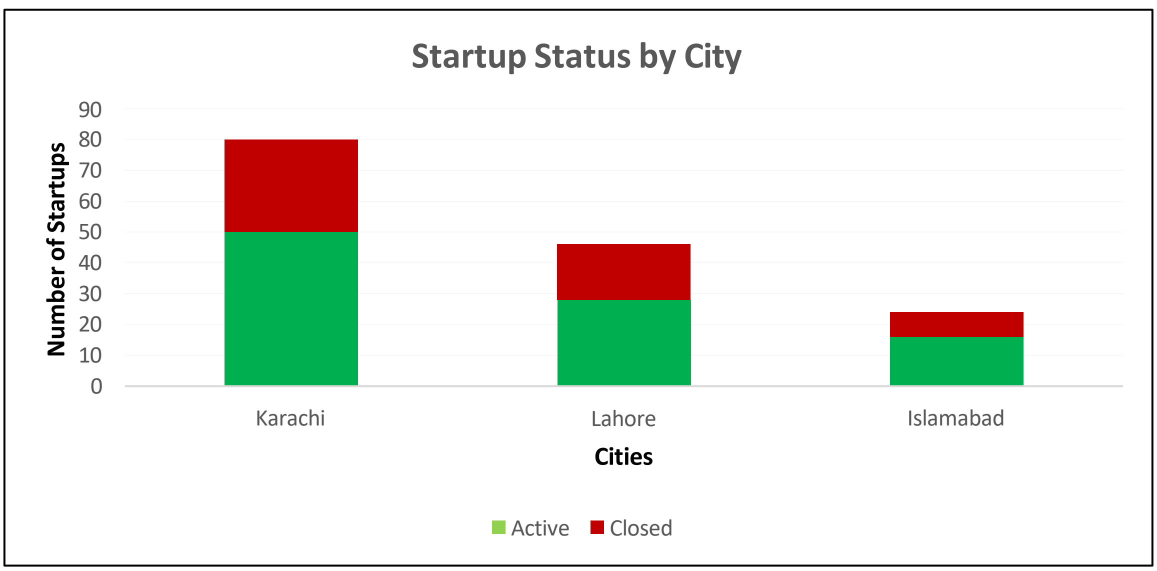


Figure 3.2: Startup Status by City

Distribution of Active and Closed in Karachi, Lahore, and Islamabad is shown in Figure 3.2. Out of the total startups, Karachi has the most startups numbering 80, while Lahore and Islamabad have 50 and 26 respectively. However, what is surprising is that while Karachi has a survival rate of 63%, and Lahore 56%, Islamabad has the highest rate at 69%.

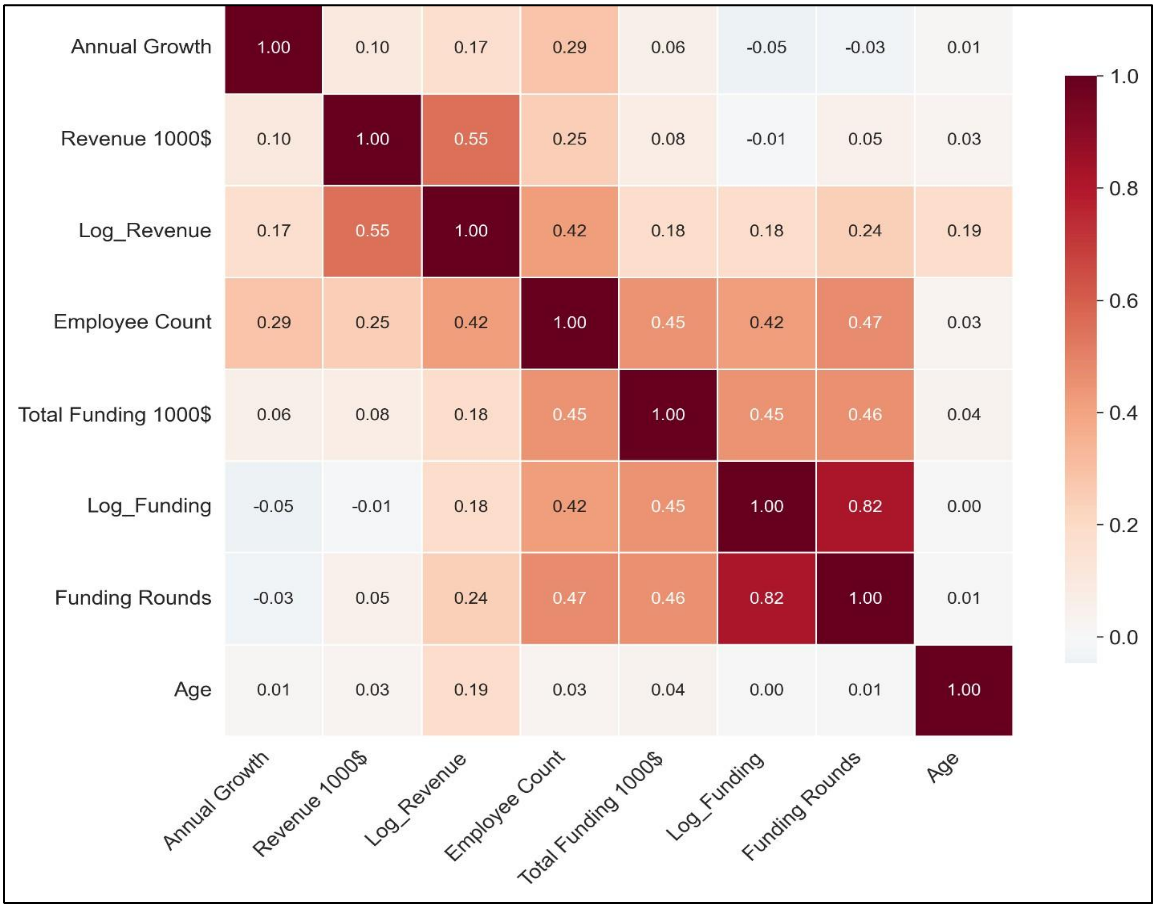


Figure 3.3: Correlation Matrix of Numeric Variables

Figure 3.3 has this correlation heatmap that shows how all the continuous variables in the study connect to each other. It uses Pearson coefficients going from negative one for strong negative links up to positive one for strong positive links and the colors get darker or lighter based on that along with the actual numbers right there. The correlation between financing rounds and log funding stands out a lot. Its r value is 0.82 which is strong and positive, that makes sense because they are basically looking at similar ideas around money coming in Employee count ties into this too. It has positive correlation with funding rounds at r equals 0.47 and then log funding at 0.42 and log revenue also 0.42. So larger teams seem to get more funding and bring in more revenue it looks like. Annual growth does not connect much with the other stuff. It is weak across the board which suggests that growth is distinctly constructed and can't be captured by funding or revenue alone. Age barely links to anything here. That matches what other studies which say startups age does not really predict the performance significantly.

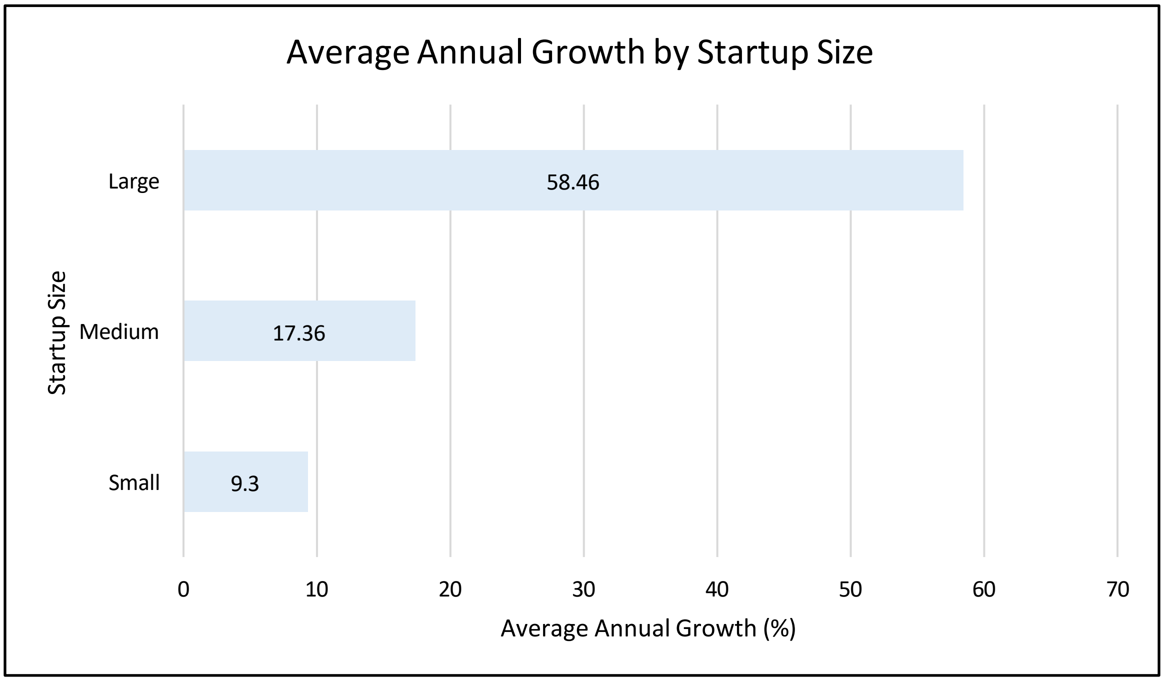


Figure 3.4: Average Annual Growth by Startup size

Figure 3.4 shows the mean annual growth by startup size. Large startups recorded a mean growth rate of 58.46%, medium-sized companies have a growth rate of 17.36%, and small startups have mean growth rate of 9.30%. These findings confirm the ANOVA findings, showing differences in the growth rates between different sizes of companies.

3.6 Statistical Methods

This section describes the statistical methods used in the study. Each method is selected to answer a specific research question.

3.6.1 Distributional Analysis (EasyFit Modeling)

To identify the underlying probability distributions of the main continuous variables, distribution fitting is performed using EasyFit software. Assessing distributional behavior is important for understanding tail behavior, skewness, and the suitability of parametric techniques. Multiple theoretical distributions are fitted to each variable, and their suitability is evaluated using goodness-of-fit tests.

For each distribution, the following hypotheses are tested:

H_0: The data follow the specified theoretical distribution.

H_1: The data do not follow the specified theoretical distribution.

Statistical significance is evaluated at predefined levels ($\alpha = 0.05$ and $\alpha = 0.01$, depending on the variable). When the null hypothesis could not be rejected, the distribution is considered an adequate representation of the data. Estimated parameters for these distributions included shape, scale, and location parameters, depending on the functional form of the distribution.

3.6.2 Analysis of Variance (ANOVA)

To determine whether continuous variables differ significantly across categorical groups, one-way ANOVA is performed. This method tests the null hypothesis of the equality of means of a dependent variable for each category of a categorical independent variable against the alternative hypothesis indicating a significant difference between at least one group's mean.

ANOVA divides the overall variation present in the data set into two components: variance between groups (explained by categorical factor), and variance within groups (due to random variations). The value of F is computed as a ratio of the mean square between groups and the mean square within groups. Large F-value and small p-value indicate statistical significance between the group means. The underlying assumptions of ANOVA, including independence of observations, normality of residuals, and homogeneity of variances have been verified prior to the analysis.

3.6.3 Independent Samples T-Test

The independent samples t-test is used to compare the means of annual growth for active and inactive startups. First, the information has been categorized into two groups; those startups who received zero funding and those having some funding amount, with a view to examining whether there is a difference in the association between growth and survival depending on the availability of funds. Then, the annual growth for active and inactive startups has been compared through t-tests separately. The Levene test is done to ascertain whether the variances of both groups are homogeneous before conducting the t-test. Effect sizes (Cohen's d) are also calculated to examine the significance of the observed differences.

3.6.4 MANOVA

A multivariate analysis of variance is conducted to find out if there are any simultaneous changes in annual growth and log revenue according to startup status (whether active or not) and by size (small, medium, and big startups). MANOVA extends ANOVA by making it capable of dealing with multiple dependent variables at one time while reducing the risk of committing Type I error. Wilks' Lambda is used as the test statistic, and partial eta squared is reported as the measure of effect size.

3.6.5 Linear Regression

The independent variables associated with startup revenue are derived from linear regression analysis. Log_Revenue is the natural log of revenue that serves as the dependent variable. Employee count, size, and time are considered independent variables. There is a heavy skewness in the initial revenue figures; therefore, it necessitated the use of the log function. A series of assumptions for linear regression has been checked before the results could be analyzed. These included no multicollinearity, homoscedasticity, errors independence, and normally distributed residuals. Multicollinearity is assessed using variance inflation factor, with VIF values below five are acceptable.

3.6.6 Cox Proportional Hazards Regression

Cox Regression is used to identify the time taken for a startup to close. While logistic regression predicts whether an event will occur, Cox regression estimates the time when it happens. This difference is important because the closure of a firm after one year is not equivalent to its closure after seven years. Time variable referred to the number of years elapsed between the start of the analysis period and 2025. The status variable indicated whether the startup is closed (1) or it remained active (0). If the startups are still active at the time of study, they are considered as they haven’t experienced the event of interest yet.

3.6.7 Kaplan-Meier Survival Analysis

To complement the results obtained from the Cox regression model, Kaplan Meier Survival Analysis is also used. The method helps in comparing different categories and makes it easier for us to know the changing nature of survival chances over time. Here, this method is used for comparison between companies showing positive growth and companies showing negative growth. It helped in establishing the statistical difference in the survival rate of both categories through the log-rank test. Furthermore, hazard functions and life tables are used for observing the maximum risk and survival probability, respectively.

3.6.8 K-Means Cluster Analysis

The K-Means Clustering Technique to form clusters based on the characteristics of the various groups of firms. Four different factors are analyzed, which includes annual growth, employee count, funding efficiency, and revenue. All the factors are standardized using the Z-score transformation method as to ensure that each factor played an equal role in forming the clusters. Testing is conducted for cluster formations involving two, three, and four clusters in order to establish the number of clusters. It turned out that three clusters gave more meaningful results.

3.7 Data Analysis Software

The Easyfit software and IBM SPSS Statistics are used for the performance of all types of statistical analysis. Data preparation, variable transformation, descriptive statistics, and all inferential tests are conducted within the SPSS environment.

3.8 Ethical Considerations

Publicly available secondary data is used in this research study. There are no personal identifiers of any particular investors, employees, and founders in the data set analyzed. This research project is done solely for academic reasons as part of BS Honors degree in Statistics.

3.9 Limitations of the Methodology

There are several limitations to this research that need to be taken into consideration. Firstly, the researcher has no influence on the process of data collection and recording when it comes to secondary data. Secondly, the findings cannot be generalized to other countries as the sample only consists of startups from Pakistan, covering its main three cities. Thirdly, the study is cross-sectional as the researcher does not examine startups through time but analyzes only one moment in their life. It is thus harder to determine the relationship between the dependent and independent variable. Fourthly, the study lacked qualitative data that would help understand the importance of certain aspects, e.g., founders' interviews. Fifthly, the models did not control for any external factors such as changes in government policies, economic crises, and political instability. Finally, there are fewer startups closed down than currently functioning in the sample, reflecting difficulty of obtaining failed startups. Survival analysis models allow dealing with such problems of imbalanced data.

CHAPTER 4

DATA ANALYSIS AND DISCUSSION

4.1 Distribution Fitting Using EasyFit

EasyFit software is used for distribution fitting in order to determine the underlying probability distributions of the important continuous variables. In each of the proposed distributions, the following hypotheses are tested:

H_0 (Null Hypothesis): The data follow the specified theoretical distribution.

H_1 (Alternative Hypothesis): The data do not follow the specified theoretical distribution.

The Kolmogorov-Smirnov, Anderson-Darling and Chi-Squared tests are applied to test the goodness-of-fit at the predetermined levels of significance (0.01 and 0.05). When the null hypothesis could not be rejected, it means that the distribution is a good representation of the data. The parameters of accepted distributions are estimated, such as shape, scale, and location, based on the functional formation of the distribution in question.

4.1.1 Distribution Fitting Results for Revenue

To identify the underlying probability distribution of revenue, four distributions are fitted. Kolmogorov-Smirnov at the 0.01 significance level is used to assess the goodness-of-fit. As per the null hypothesis (H_o) the data are distributed as per the specified theoretical distribution.

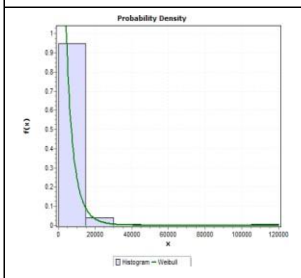
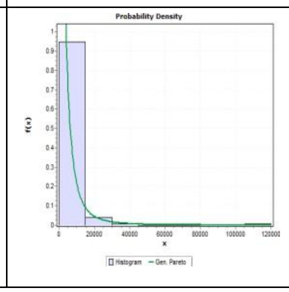
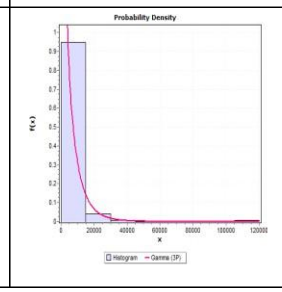
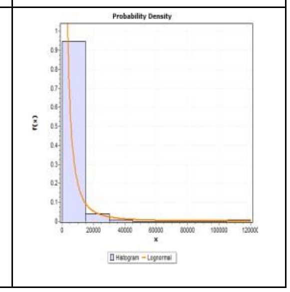
Table 4.1.1: Kolmogorov–Smirnov Goodness-of-Fit Test for Revenue (n = 150)

Distribution	Test Statistic	Critical Value (0.01)	Decision ($\alpha = 0.01$)
Weibull	0.07711	0.13309	Do Not Reject H_o
Gen. Pareto	0.07895	0.13309	Do Not Reject H_o
Gamma (3P)	0.11582	0.13309	Do Not Reject H_o
Lognormal	0.08694	0.13309	Do Not Reject H_o

Table 4.1.2: Estimated Distribution Parameters for Revenue

Distribution	Shape Parameter	Scale Parameter	Location Parameter
Weibull	α = 0.81912	β = 3292.6	—
Gen. Pareto	k = 0.44687	σ = 2584.0	μ = -78.172
Gamma (3P)	α = 0.57571	β = 7988.3	γ = 35.0
Lognormal	σ = 1.5128	μ = 7.4388	—

Table 4.1.3: Histogram of Revenue with Fitted Theoretical Distributions

Weibull	**Gen. Pareto**	**Gamma (3P)**	**Lognormal**
			

The Kolmogorov-Smirnov test (α= 0.01) is applied to four possible revenue distributions. The rejection of all the distributions is not made since the test statistics are less than the critical value of 0.13309. Weibull distribution is the closest (statistic = 0.07711), then Generalized Pareto (0.07895), Lognormal (0.08694) and Gamma (3P) (0.11582). The estimated parameters satisfy the skewness and tail properties: Weibull shape (α = 0.81912 < 1) implies decreasing probability of extreme values; Lognormal parameters (σ = 1.5128, μ = 7.4388) imply that the skewness is to the right, and Generalized Pareto location (μ = -78.172) confirms shifted heavy-tailed behavior. These findings validate the application of logarithmic transformation on revenue in the regression analyses.

4.1.2 Distribution Fitting Results for Total Funding

To identify the underlying probability distribution of Total Funding, four distributions are fitted. Kolmogorov-Smirnov at the 0.05 significance level is used to assess the goodness-of-fit. As per the null hypothesis (H_o) the data are distributed as per the specified theoretical distribution.

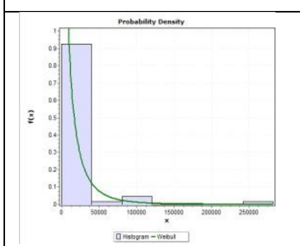
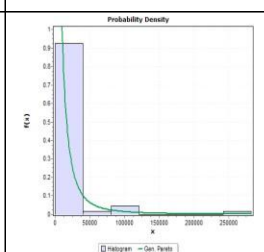
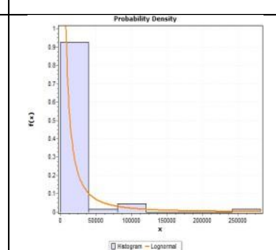
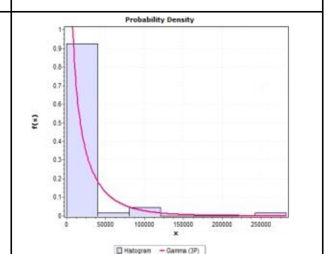
Table 4.1.4: Kolmogorov–Smirnov Goodness-of-Fit Test for Total Funding

Distribution	Test Statistic	Critical Value (0.05)	Decision (α = 0.05)
Weibull	0.05428	0.16615	Do Not Reject H_0
Pareto	0.06942	0.16615	Do Not Reject H_0
Lognormal	0.08930	0.16615	Do Not Reject H_0
Gamma	0.14846	0.16615	Do Not Reject H_0

Table 4.1.5: Estimated Distribution Parameters for Total Funding

Distribution	Shape Parameter	Scale Parameter	Location Parameter(s)
Weibull	α = 0.63599	β = 8945.8	—
Pareto	k = 0.60396	σ = 6779.3	μ = -578.58
Lognormal	σ = 1.903	μ = 8.2927	—
Gamma	α = 0.4324	β = 41701.0	γ = 35.0

Table 4.1.6: Histogram of Total Funding with Fitted Theoretical Distributions

Weibull	**Pareto**	**Lognormal**	**Gamma**
			

The Kolmogorov–Smirnov test (α = 0.05) is applied to four candidate distributions for total funding. None of the distributions are rejected as all the test statistics is less than the critical value of 0.16615. Weibull distribution provided the best fit (statistic = 0.05428) and the next are Generalized Pareto (0.06942), Lognormal (0.08930) and Gamma (3P) (0.14846). Extreme positive skewness and heavy-tailed behavior are confirmed by the predicted parameters. The shape parameter of the Weibull (α = 0.63599 < 1) indicates a decreasing value of the hazard rate, and therefore most of the funding values are concentrated around zero with a small number of very large amounts. The Lognormal parameters (σ = 1.903, μ = 8.2927) confirm right skew. Shifted heavy-tailed behavior is confirmed by the

Generalized Pareto shape parameter (k = 0.60396) and negative location parameter (μ = -578.58). The Gamma shape parameter ($\alpha = 0.4324 < 1$) is another confirmation of the rapid decline in density. These findings support the use of the logarithmic transformation of total funds in analyses.

4.2 Analysis of Variance (ANOVA)

One-way Analysis of Variance (ANOVA) is used to determine whether continuous variables differ significantly across categorical groups. The differences in the size of startups (Small, Medium, and Large) are compared in terms of revenue, total capital and annual growth. In each test the significance level is set at $\alpha = 0.05$.

4.2.1 Analysis of Variance (ANOVA) by Startup Size

The determination of the difference in revenue, total funding, and annual growth across the categories of startup size (Small = 1, Medium = 2, Large = 3) is done using a one-way Analysis of Variance. The null hypothesis (H_o) states that the means of the dependent variable are same in all groups of sizes against the alternative hypothesis (H_1) which states that at least one group mean differs significantly.

Table 4.2.1: ANOVA Table for Revenue, Funding, and Annual Growth by Startup Size

Variable	Sum of Squares	df	Mean Square	F	Sig.	Decision (α=0.05)
Between Size and Revenue	3,499,437,594.80	2	1,749,718,797.40	18.819	< 0.001	Reject H_o
Between Size and Total Funding	10,837,744,999.83	2	5,418,872,499.92	7.817	0.001	Reject H_o
Between Size and Annual Growth	20,231.38	2	10,115.69	5.371	0.006	Reject H_o

All the three dependent variables are found to be significantly different among the groups in terms of startup size at the 5% level of significance. Revenue exhibited the biggest impact ($F = 18.819$, $p < 0.001$), which is followed by total funding ($F = 7.817$, $p = 0.001$) and annual growth ($F = 5.371$, $p = 0.006$). On the basis of these findings, the size of startups is a meaningful factor in distinguishing performance either financial or operational. Higher revenue and funding are frequently obtained by larger firms, and the rates of growth vary widely depending on the size.

4.2.2 Analysis of Variance (ANOVA) by Industry

The Analysis of Variance is used to ascertain whether there is a significant difference in revenue, total funding and funding rounds among different industry groups. The alternative hypothesis (H_1) implies that there is a significant difference between at least one of the group means, the null hypothesis (H_0) implies that they are equal across all industry groups.

Table 4.2.2: ANOVA Table for Revenue, Funding, and Funding Rounds by Industry

Variable	Sum of Squares	df	Mean Square	F	Sig.	Decision
Between Industry and Revenue	4,826,814,080.18	11	438,801,280.02	4.907	< 0.001	At $\alpha = 0.05$ Reject H_0
Between Industry and Total Funding	13,936,084,992.32	11	1,266,916,817.48	1.769	0.065	At $\alpha = 0.10$ Reject H_0
Between Industry and Funding Rounds	39.60	11	3.60	3.142	0.001	At $\alpha = 0.05$ Reject H_0

In this ANOVA table, the difference in revenue among the industry categories is also found to be significant ($F(11, 138) = 4.907$, $p < 0.001$), indicating that some of the industries generate a considerably high revenue as compared to the rest. At $\alpha= 0.05$ ($p = 0.065$), there is no significant value at the total funding. But the null hypothesis is rejected at the 10% level ($\alpha= 0.10$) which means that there is little evidence of industry based differences in funding level.The number of funding rounds varies by industry, as seen by the substantial differences in funding rounds between industry categories ($F(11, 138) = 3.142$, $p = 0.001$).

Overall, industry affiliation impacts investment rounds and revenue more than total funding. This pattern holds for both funded and self-funded companies.

4.3 Independent Samples T-Test: Annual Growth by Funding Status

The data is divided into two groups based on funding status: startups with zero funding and startups with some funding in order to find out if annual growth varies between active and closed startups. An independent samples t-test is used to compare the growth of each group on an annual basis, the active (coded 1) and the closed startups (coded 2).

4.3.1 Zero Funded Startups

The mean growth of startups that did not receive any external funding ($n = 83$) is 27.78% per year with active startups ($n = 67$) and -16.54% per year with closed startups ($n = 16$). The 44.33% difference is statistically significant ($t(81) = 4.388$, $p < 0.001$). Equal variances are found using Levene's test ($F = 0.220$, $p = 0.641$). There is a significant practical difference between the two groups, as indicated by the big effect size (Cohen's $d = 1.22$).

Table 4.3.1: Independent Samples T-Test for Zero-Funded Startups

Status	N	Mean (%)	Std. Deviation	t	df	p-value	Cohen's d
Active	67	27.78	38.64	4.388	81	< 0.001	1.22
Closed	16	-16.54	23.41				

4.3.2 Funded Startups

Active companies ($n = 27$) have a mean annual growth of 45.61% while closed startups ($n = 40$) have a mean annual growth of -14.38% among startups that received some external funding ($n = 67$). The 59.99% difference is statistically significant ($t(65) = 6.033$, $p < 0.001$). The equal variances not assumed row is used since Levene's test revealed uneven variances ($F = 10.682$, $p = 0.002$). The effect size, $d = 1.50$ by Cohen, is very large, and indicates a very practical difference.

Table 4.3.2: Independent Samples T-Test for Funded Startups

Status	N	Mean (%)	Std. Deviation	t	df	p-value	Cohen's d
Active	27	45.61	55.63	6.033	65	< 0.001	1.50
Closed	40	-14.38	24.35				

Startups that remained active grew much more each year compared to those that closed, whether they had funding or not. The difference in growth is bigger among funded startups at 59.99%, while zero-funded ones showed a 44.33% gap. Annual growth really sets apart those that survive from those that fail, with strong impact seen in both cases ($d = 1.22$ and $d = 1.50$). Interestingly, active startups keep growing positively at rates of 27.78% and 45.61%. On the other hand, closed startups see negative average growth, with declines of -16.54% and -14.38%. This pattern is for both funded and self-funded companies, highlighting that negative growth often serves as a main reason of failure.

4.4 MANOVA Results

A multivariate analysis of variance is conducted to identify the variation in annual growth and log revenue among the sizes of startups (small, medium, and large) and business condition (active vs. closed). The descriptive statistics indicated that the larger startups are more likely to have higher growth and revenues compared to the smaller ones and active startups always have higher annual growth compared to closed startups of all size groups.

Table 4.4.1 Multivariate Test

Effect	Wilks' Lambda	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Intercept	0.065	1016.621	2	142	< 0.001	0.935
Status	0.856	11.923	2	142	< 0.001	0.144
Size	0.776	9.604	3	284	< 0.001	0.119

As the mean values of the dependent variables are not zero, the intercept is significant ($p < 0.001$). The startup status has a strong influence on the combined dependent variables (Wilks' $\Lambda = 0.856$, $p < 0.001$). There is a significant effect (partial $\eta^2 = 0.144$). The startup size also has a strong influence on the combined dependent variables (Wilks' $\Lambda = 0.776$, $p < 0.001$). The effect size is medium (partial $\eta^2 = 0.119$)

Table 4.4.2 Tests of Between-Subjects Effects

Source	Dependent Variable	F	Sig.	Partial Eta Squared	Decision
Status	Annual Growth	23.926	< 0.001	0.143	Significant
Status	Revenue	0.003	0.953	0.000	Not Significant
Size	Annual Growth	7.129	0.001	0.091	Significant
Size	Revenue	14.697	< 0.001	0.171	Significant

In the analysis of each of the dependent variables separately, startup status has a significant effect on annual growth ($F = 23.926$, $p < 0.001$), but not on revenue ($p = 0.953$). This means that the difference between active and closed startups is characterized by annual growth not revenue. Both annual growth ($F = 7.129$, $p = 0.001$) and revenue ($F = 14.697$, $p < 0.001$) are significantly affected by startup size. It shows that bigger startups are more successful in attracting more revenue and growth than smaller ones. These findings confirm the importance of the size and status of a startup in the distinction of startup performance. Although the growth and the increase in revenue differ among the different categories of sizes, growth is the factor that differentiates closed firms and active ones.

4.5 Linear Regression

4.5.1 Revenue Transformation

Preliminary analysis of the revenue variable revealed significant deviation from normality. The skewness of 9.009 and kurtosis of 101.105 indicated a highly positive skewness and the Kolmogorov-Smirnov test is statistically significant ($p < 0.001$). This trend is common when a small number of high performing startups generate large revenue while majority of the startups generate low revenue. To correct this violation of the normality assumption, the revenue variable is adjusted using the natural logarithmic form. Log_Revenue = LN(Revenue_1000$ + 1) is the formula used to calculate the transformed variable.

This transformation fulfilled three purposes: (1) the distribution of the dependent variable is normalized, (2) the effect of extreme outliers is reduced, and (3) percent-based interpretation of coefficients, which is more important in business contexts, is achieved.

So that the log revenue is used as dependent variable for all further regression analyses. The coefficients in the model are expressed as a percentage change in the revenue, given a one-unit change in the predictor variable.

4.5.2 Model Setup

Table 4.5.1 Linear Regression Results

Sr #	Hypothesis	Test Statistic	P-value	Result
1	H_1: Employee Count has a significant effect on Revenue	F = 6.732	0.010	Significant
2	H_1: Startup Size has a significant effect on Revenue	F = 11.157	0.001	Significant
3	H_1: Time has a significant effect on Revenue	F = 4.484	0.036	Significant

To find the variables related to startup revenue, a linear regression analysis is performed. The dependent variable is the revenue, and the independent variables are Time, size, and Count of Employees. The regression model is statistically significant ($F(3, 146) = 17.200$, $p < 0.001$) indicating that the predictors combined explain a large portion of the variation in the startup revenue. 26.1% of the revenue variance is explained by the model ($R^2 = 0.261$). The Durbin-Watson Statistics is 1.892 which is close to 2, indicating that there is no significant autocorrelation in the residuals

Table 4.5.2: Regression Coefficients

Model	Unstandardized B	Std. Error	Standardized β	t	Sig.
Constant	5.503	0.417	—	13.209	< 0.001
Employee Count	0.004	0.001	0.232	2.595	0.010
Startup Size	0.720	0.216	0.299	3.341	0.001
Time	0.096	0.045	0.152	2.118	0.036

Linear Regression equation:

Revenue = 5.503 + 0.004(Employee Count) + 0.720(Startup Size) + 0.096(time)

4.5.3 Predicted Values Calculation

The predicted Log_Revenue is computed for various combinations of predictor values, as shown in the table below. When all predictors are zero, the baseline revenue is represented by the constant term (5.503). Every predictor adds its coefficient multiplied by the value of the predictor.

Table 4.5.3: Calculation of Predicted Revenue

Constant	Employee Count	Startup Size	Time	Predicted Revenue
5.503	0	0	0	5.503
5.503	0.004 × 1 = 0.004	0	0	5.507
5.503	0	0.720 × 1 = 0.720	0	6.223
5.503	0	0	0.096 × 1 = 0.096	5.599
5.503	0.004 × 1 = 0.004	0.720 × 1 = 0.720	0.096 × 1 = 0.096	6.323

Multicollinearity Diagnostics

To make sure the predictor variables are independent, multicollinearity diagnostics is carried out. All the values of Variance Inflation Factor (VIF) are much less than the accepted threshold of 5 and are between 1.011 and 1.587. The tolerance values are between 0.630 and 0.989. The highest condition index is 8.805 with all the rest being less than 10. These results indicate that there is no problem of multicollinearity and all predictors contribute independently to the model.

Table 4.5.4: Multicollinearity Diagnostics

Predictor	Tolerance	VIF
Employee Count	0.636	1.573
Enterprise Tiers	0.630	1.587
Time	0.989	1.011

The size of the company is determined as the best predictor of revenue. Larger startups generated substantially higher revenues, with each increase in size category associated with a 72.0% increase in revenue. Also, the growth in number of employees positively influence revenue, with each additional employee adding 0.4 percent to the revenue. Moreover, older startups did better than younger, and revenue growth increased by 9.6 percent per additional year of operation. The multicollinearity tests confirmed that the predictors do not have any issues, and the model explained 26.1% of the variation in revenue.

4.6 Cox Regression

Cox Proportional Hazards Regression is used to analyze time-to-event data. Cox regression predicts the time when a startupwill fail unlike the logistic regression that only predicts whether the startup will fail. It also deals with censored data (still-operating startups at the end of the study period) in an appropriate manner.

4.6.1 Model Setup

A Cox Regression is employed to identify the variables that are associated with the time it takes to close a startup. The dependent variable is time to closure measured in years (Age). The model included 3 predictors:

Negative Growth: 1 = negative growth, 0 = positive growth

Revenue: LN(Revenue_1000$ + 1)

Total Funding: Total funding raised (in $1000)

Table 4.6.1: Summary of Cox Regression Results

Sr #	Hypothesis	Test Statistic	P-value	Result
1	H_1: Negative Growth increases the hazard of closure	Wald $\chi^2 = 47.986$	< 0.001	Significant
2	H_1: Revenue decreases the hazard of closure	Wald $\chi^2 = 5.314$	0.021	Significant
3	H_1: Total Funding increases the hazard of closure	Wald $\chi^2 = 4.763$	0.029	Significant

The entire model is statistically significant with a $\chi^2(3) = 80.926$, $p = 0.001$ indicating that the combination of these factors explains the time to startup closure. The -2 Log Likelihood decreased from 481.627 (null model) to 400.702 (final model), demonstrating improved model fit.

4.6.2 Regression Coefficients

All three predictors are statistically significant at the 0.05 level.

Table 4.6.2: Variables in the Equation

Predictor	B	SE	Wald	df	Sig.	Exp(B)
Negative_Growth	2.719	0.393	47.986	1	< 0.001	15.165
Revenue	-0.288	0.125	5.314	1	0.021	0.750
Total Funding ($1000)	0.003	0.0001	4.763	1	0.029	1.000

Cox Regression Equation

Generally, the Cox Proportional Hazards model is expressed as:

$$\mathbf{h(t) = h_0(t) \times exp\ (\beta_1X_1 + \beta_2X_2 + ... + \beta_kX_k)}$$

Where:

$h(t)$ = Hazard function at time t (risk of closure at a given time)

$h_o(t)$ = Baseline hazard function (hazard when all predictors = 0)

exp = Exponential Function

$\beta_1, \beta_2...$ = Regression coefficients

$X_1, X_2...$ = Predictor variables

Based on the regression results, the Cox model for predicting startup closure is:

$$\mathbf{h(t) = h_0(t) \times exp\ [2.719(Negative\_Growth) - 0.288(Revenue) + 0.003(Total\ Funding)]}$$

4.7 Kaplan-Meier Survival Analysis

To complement the Cox Regression results, Kaplan-Meier survival analysis is conducted to compare startups with positive growth versus startups with negative growth. Among 101 positive growth startups only 8 closures occurred (92.1% censored). Out of the 49 startups that had negative growth, 48 closures occurred (2.0% censored).

The log-rank test (χ^2 = 90.162, df = 1, p < 0.001) is used to confirm the statistical significance of the survival difference between the two groups. The median survival of negative growth startups is 7.0 years (95% CI: 6.51 – 7.49 years) whereas positive growth companies did not hit the median survival duration in the period of observation.

Table 4.7.1: Life Table for Positive Growth Startups

Time (Years)	Remaining Cases	Cumulative Events	Survival Probability
5	92	1	0.989
6	75	3	0.963
7	52	4	0.945
8	37	7	0.870
9	24	8	0.836

Note: The table includes intervals where events occurred. No further events are observed beyond year 9 for positive growth startups

Table 4.7.1 shows positive growth startups have consistently high survival, with 83.6% still active after 9 years and no further closures beyond that point.

Table 4.7.2: Life Table for Negative Growth Startups

Time (Years)	Remaining Cases	Cumulative Events	Survival Probability
4	48	1	0.980
5	40	10	0.796
6	29	21	0.571
7	15	34	0.306
8	7	41	0.153
9	5	43	0.109
10	3	45	0.066
11	2	46	0.044
12	1	47	0.022
13	0	48	0.000

Table 4.7.2 shows startups with negative growth did not last long. Over time, their survival rate drastically decreased. Merely 30.6% of them survived for more than seven years. That percentage dropped to just 6.6% by the tenth year. And none of these startups remained in operation after 13 years.

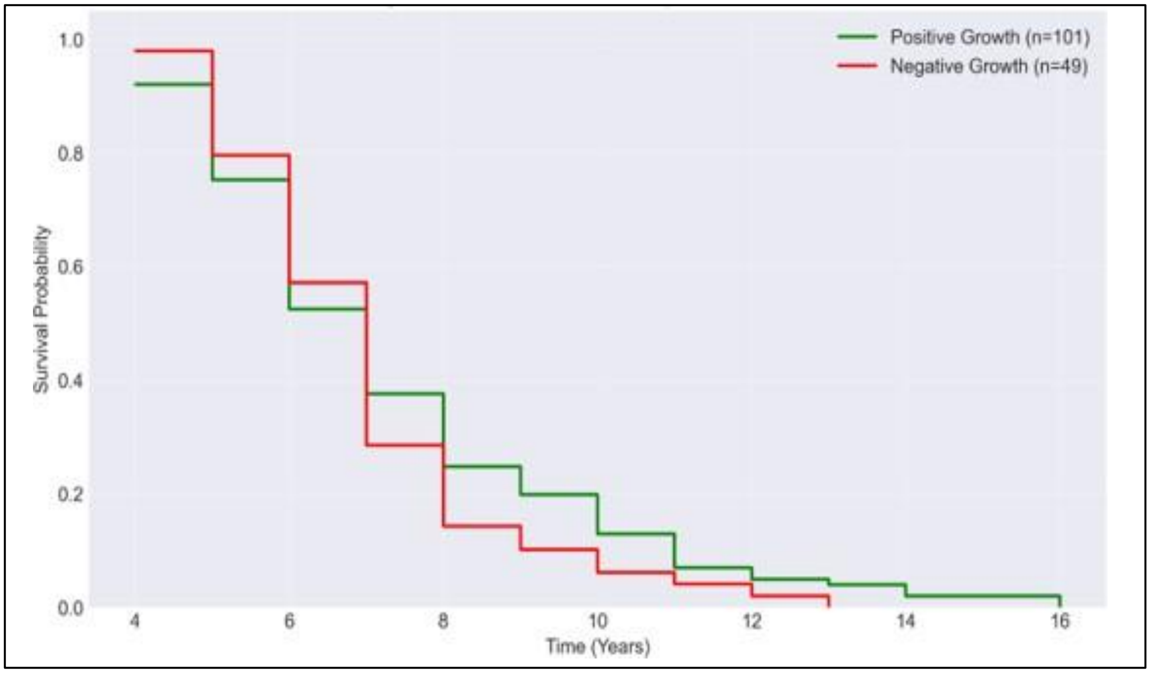


Figure 4.7.1: Kaplan-Meier Survival Curves by Growth Status

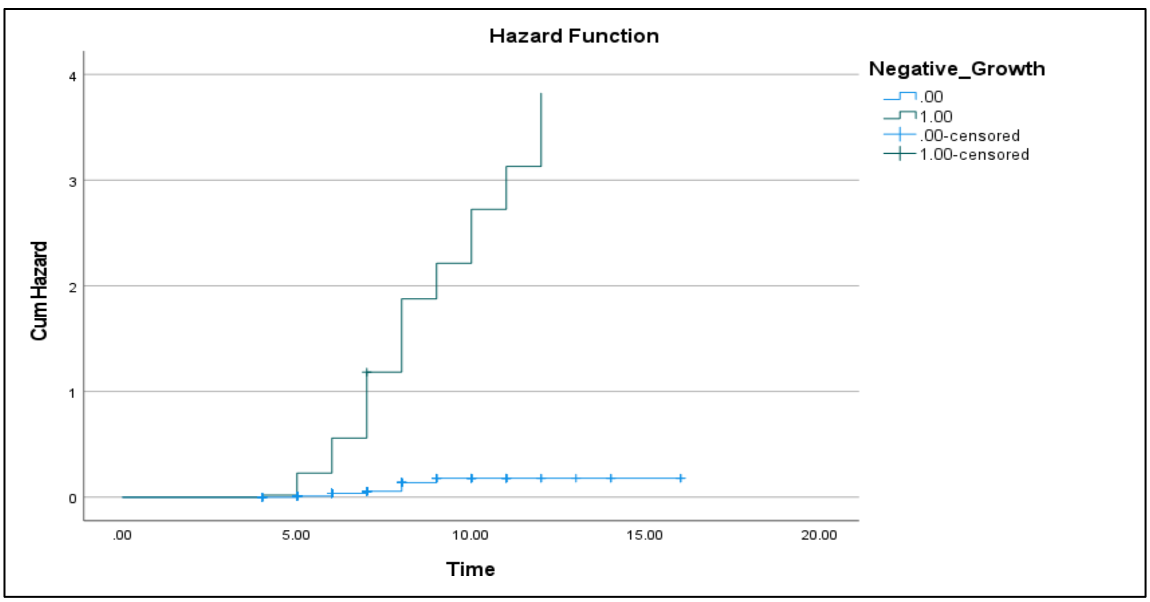


Figure 4.7.2: Cumulative Hazard Function by Growth Status

Throughout the observation period, startups showing good growth have high survival. Negative growth startups, on the other hand, experienced a decline. Only 30.6% of negative growth startups survived after seven years and none after thirteen years. The hazard function shows that the risk of shutdown of startups experiencing negative growth is concentrated in the years five to eight. These results are consistent with the Cox Regression results, which showed that the Growth variable (negative growth coded as 1) is the best predictor of closure hazard (Exp(B) = 15.165, $p < 0.001$).

4.8 K-Means Cluster Analysis

K-means cluster analysis is used to identify startup groups according to their characteristics. Four variables are included in the analysis: annual growth, employee count, funding efficiency, and revenue. To guarantee equal weighting, all variables are standardized to Z-scores before analysis. The analysis is carried out in two stages due to the existence of an extreme outlier (ASSYST APPAREL SOLUTIONS with revenue of $120,000K) that forms a single-member cluster in the entire sample. The results are first shown for the entire sample, including the outlier. To get a more representative segmentation, the analysis is then repeated on the remaining 149 startups after the outlier is eliminated.

4.8.1 Cluster Results Including Outlier (Full Sample, n = 150)

The full sample analysis yields three clusters. These clusters are described in the following tables.

Table 4.8.1: K-Means Cluster Results

Variable	Cluster 1 (Outlier) n=1	Cluster 2 n=136	Cluster 3 n=13	F-value	p-value
Revenue	10.752	-0.141	0.648	360.616	**< 0.001**
Funding Efficiency	11.238	-0.072	-0.116	428.068	**< 0.001**
Employee Count	-0.077	-0.251	2.634	144.565	**< 0.001**
Annual Growth	-0.068	-0.128	1.344	15.339	**< 0.001**

Table 4.8.2: Cluster Distribution by Startup Status

Cluster	n	Active (n)	Active (%)	Closed (n)	Closed (%)
Cluster 1 (Outlier)	1	1	100.0%	0	0.0%
Cluster 2	136	84	61.8%	52	38.2%
Cluster 3	13	9	69.2%	4	30.8%
Total	150	94	62.7%	56	37.3%

Table 4.8.3: Cluster Profiles – Descriptive Statistics by Cluster

Variable	Cluster 1 (Outlier) n=1	Cluster 2 n=136	Cluster 3 n=13
Revenue ($1000)	120,000.00	3,079.51 (±3,763.60)	11,553.85 (±9,102.44)
Annual Growth (%)	12.00	9.30 (±35.01)	75.05 (±81.81)
Employee Count	46.00	30.09 (±32.38)	293.00 (±151.52)
Funding Efficiency	120,000.00	2,084.68 (±3,816.61)	1,616.17 (±5,824.12)

Values are mean (standard deviation).

The full sample reveals three clusters. However, Cluster 1 contains only one startup, which is statistically insufficient to represent a meaningful segment. Therefore, the outlier is removed and the analysis is repeated.

4.8.2 Cluster Results Without Outlier (Refined Sample, n = 149)

After removing the outlier, K-means cluster analysis is repeated on the remaining 149 startups.

Table 4.8.4: K-Means Cluster Results (n = 149)

Variable	Cluster 1 (n=13)	Cluster 2 (n=136)	F-value	p-value
Revenue	0.648	-0.141	43.098	< 0.001
Funding Efficiency	-0.116	-0.072	0.161	0.689
Employee Count	2.634	-0.251	289.113	< 0.001
Annual Growth	1.344	-0.128	30.673	< 0.001

Table 4.8.5: Cluster Distribution by Startup Status (n = 149)

Cluster	n	Active (n)	Active (%)	Closed (n)	Closed (%)
Cluster 1 (High Growth)	13	9	69.2%	4	30.8%
Cluster 2 (Mainstream)	136	84	61.8%	52	38.2%
Total	149	93	62.4%	56	37.6%

Table 4.8.6: Cluster Profiles – Descriptive Statistics (n = 149)

Variable	Cluster 1 n=13	Cluster 2 n=136
Revenue ($1000)	11,553.85 (9,102.44)	3,079.51 (3,763.60)
Annual Growth (%)	75.05 (81.81)	9.30 (35.01)
Employee Count	293.00 (151.52)	30.09 (32.38)
Funding Efficiency	1,616.17 (5,824.12)	2,084.68 (3,816.61)

Values are mean (standard deviation).

4.8.3 Cluster Visualization

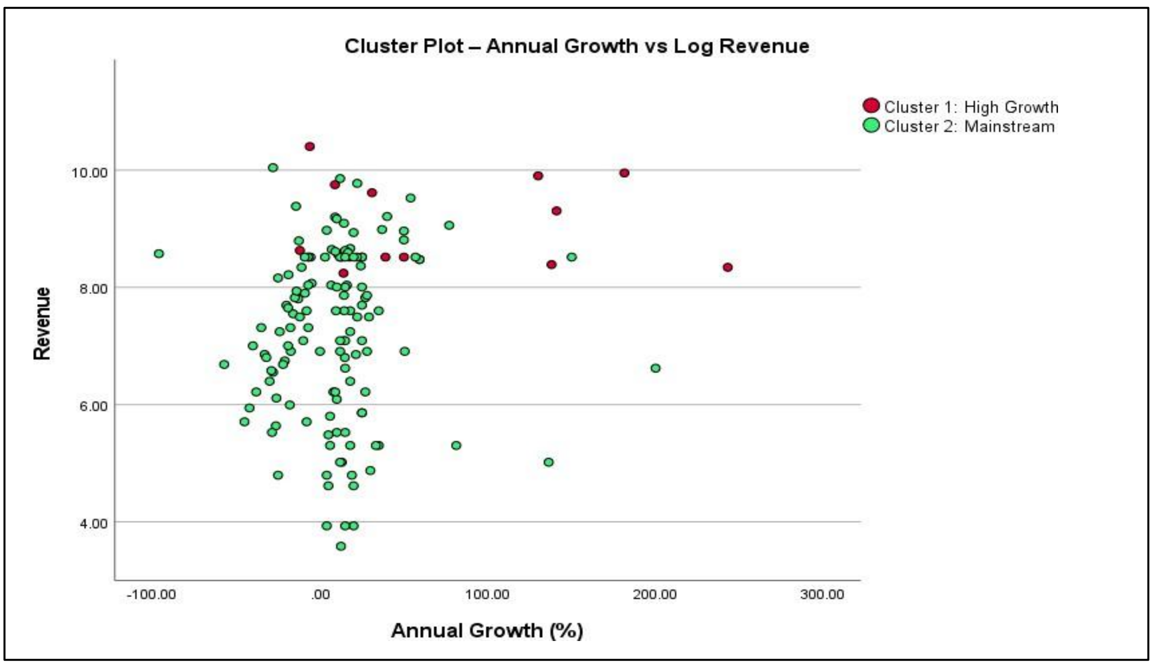


Figure 4.8.1: Annual Growth vs Revenue by Cluster

Cluster Plot is of revenue versus annual growth. Cluster 2 (Mainstream, n = 136) is displayed in green, and Cluster 1 (High Growth, n = 13) is displayed in red. Although there is considerable overlap, the figure indicates that high-growth businesses are typically located on the right side of the graph. The fact that some high-growth startups appear at lower growth values implies that the four factors: revenue, funding efficiency, employee count, and annual growth, combine to determine cluster assignment rather than growth alone. The Y-axis represents transformed log-revenue, which compresses the revenue scale and impacts the visual interpretation.

The improved analysis (without outlier) reveals two separate clusters.

i. **Cluster 1 (High Growth, n = 13, 8.7%):** These startups have above average revenue ($11,553.85K), large teams (293 workers), and high annual growth (75.05%). Compared to the mainstream cluster, their survival rate is significantly greater at 69.2%.

ii. **Cluster 2 (Mainstream, n = 136, 91.3%):** This cluster is representative of the average Pakistani startup, with modest performance in every indicator. They make up 92.9% of all closed startups and have a 61.8% survival rate.

There is a substantial difference ($p < 0.001$) in revenue, employee count, and annual growth between clusters. Once the outlier is eliminated, funding efficiency does not ($p = 0.689$), suggesting that it is not a differentiating factor. Although nearly one-third of high-growth firms fail, they have a higher survival rate than mainstream startups (69.2% vs. 61.8%). This is consistent with regression analysis, which indicates that while rapid growth does not ensure success, growth is a significant predictor of survival.

CONCLUSION

This chapter summarizes the key findings of the study. It presents the results and answers the research questions.

The analysis begins by examining the basic characteristics of the 150 Pakistani startups in the sample. The descriptive statistics reveal that revenue and total funding are highly skewed. A handful of startups generate massive amounts, while most earn quite modest revenues. The same pattern holds for funding. Some companies attract millions, while most attract no funding at all. Workforce size also varies considerably. Some startups employ hundreds of people, but most have only five employees. These findings show that there is no single type of Pakistani startup. Some are bootstrapped, and some are funded. So, log transformations are applied to revenue and funding before analyses.

EasyFit software is used to determine which probability distributions best fit the continuous data. Revenue is best fitted by the Weibull, Generalized Pareto, Gamma, and Lognormal distributions, which implies that the data are positively skewed and heavy-tailed. Total funding similarly fits the Weibull, Pareto, Gamma, and Lognormal distributions. These results justify the use of log transformations in subsequent analyses.

ANOVA is performed to test whether revenue, funding, and growth differs across startup size and industry categories. Larger startups have significantly more revenue, funding, and growth than smaller ones. Across industries, significant differences are found in revenue and funding rounds. At $\alpha = 0.10$, the Total Funding differs significantly, suggesting that industry may have a weak influence on total funding.

An independent samples t-test is conducted to compare active and closed startups. The annual growth of the successful startups is much greater than that of the failed startups. This is true both for funded and unfunded startups. The average annual growth of active zero-funded startups is 27.78% while the decrease of closed startups is 16.54%. The effect is even greater for startups that have secured funding. Closed ones grew by -14.38%, while active ones by 45.61%.

MANOVA is conducted to examine whether annual growth and revenue differ by startup size and status. The results indicated that the dependent variables, as a whole, are affected by both size and status. When broke it down, it found that status did not impact revenue but did impact growth. While the growth rates of open and closed startups are different, the revenue is not. This is a crucial realization. If a startup is not growing, it could make decent money and still go out of business. But size matters for both revenue and growth. Larger startups earned more income and grew faster. This aligned with my cluster analysis results, which indicated high-growth startups also tend to be larger.

A linear regression model is built to predict revenue. According to the linear regression model, revenue grows with more employees, larger size, and more years of operation. A startup with 100 employees, medium size, and 5 years of operation is estimated to generate millions in revenue. The model explains about 26% of the variance in revenue and is statistically significant. Startup size is the strongest predictor. Revenue increases by 72% for each increase in size category. Employee count also has a positive impact, with each additional employee adding 0.4% to revenue. Time is also important. Each additional year of operation increases revenue by about 10%. The Durbin-Watson statistic is close to 2, indicating no autocorrelation among the residuals

To determine what causes startups to shut down, Cox regression is used. This method is ideal because it models the time it takes for a startup to close, not simply whether it closes. Three predictors are included in the model. Negative growth is by far the strongest. Startups with negative growth have a 15.165 times higher hazard of closure compared to those with positive growth ($B = 2.719$, $p < 0.001$). Revenue is a protective factor. Each 1% increase in revenue reduces the hazard of closure by 25% ($B = -0.288$, $p = 0.021$). Funding has a minor effect, but it is statistically significant ($B = 0.003$, $p = 0.029$). Higher funding is associated with a slightly higher risk of closure. This finding, termed the "funding paradox," suggests that overfunding young startups may lead to careless spending, delayed organizational changes, and unhealthy growth.

Kaplan-Meier survival analysis is used to complement the Cox results. The survival curves tell a clear story. Startups with positive growth maintain high survival rates year after year. By contrast, startups with negative growth see a dramatic decline in their survival probability. Only 30% remain after seven years, and none survive beyond thirteen years. The hazard function indicates that the risk of closure for negative growth startups is highest between years five and eight. This aligns with the Cox model. Negative growth is not merely a warning sign. It is a guaranteed path to failure.

K-means cluster analysis is performed to identify how different types of startups group together. The first attempt includes an extreme outlier, a company with unusually high revenue. That outlier forms its own cluster of one, which is not useful. After its removal, the data clearly separates into two clusters. The first cluster, called High Growth, contains only 13 startups. These are the best performers in the dataset. They have large teams, above average revenue, and very fast growth. Their survival rate is 69%. The second cluster, called Mainstream, contains the remaining 136 startups. They have average revenue, small teams, and modest growth. Their survival rate is 62%. Even among high growth startups, nearly one-third still fail, showing that growth alone does not guarantee survival.

Higher growth decreases the risk of failure, and higher revenue reduces the likelihood of closure. This funding paradox is one of the most interesting findings of the study. All research questions are answered. Pakistani startups are characterized by wide variation in revenue, funding, and growth. Survival is driven by growth, not by funding. Multidimensional methods such as cluster analysis and MANOVA prove useful for uncovering patterns that simpler methods might miss.

Our findings align closely with the work of Cantamessa et al. (2018), who analyzed 214 startup failure post-mortems. They found that the absence of a structured business development strategy was a primary determinant of failure in most cases. This supports my finding that negative growth is the strongest predictor of startup closure. A startup without a clear forward route is likely to stagnate, which matches my Cox regression result that negative growth increases the hazard of closure by over fifteen times. Cantamessa et al. (2018) also noted that many failing startups lacked a clear path to scaling. This mirrors my cluster analysis finding that only 8.7% of startups belonged to a high-growth cluster, while the vast majority remained stuck in a mainstream cluster with modest growth and high failure rates. Where my study adds value is in the Pakistani context. Cantamessa et al. (2018) studied startups globally, but my research applies the same lens to an under-researched emerging economy where funding declined by 88% between 2022 and 2024. Hence growth separates survivors from failures.

RECOMMENDATIONS

- Focus on growth before seeking large funding rounds. Negative growth is a fast track to failure.
- Funding a startup without positive growth is risky. Overfunding leads to careless spending and delayed pivots.
- Startups should build sustainable business models that generate revenue without relying entirely on external funding.
- Expand the sample to multiple countries. Conduct longitudinal studies. Include qualitative interviews with founders.

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